CS-471: Machine Learning

Week 6

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Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Decision Tree Stopping Criteria
- ID3, C4.5 and CART Algorithms
- Ensemble Methods

Recap of Previous Lecture

- How do we make a Decision Tree?
- What is Gini Impurity?
- How do we find Gini Impurity of a set? Attribute?
- How do we find disorder/entropy of a set? Attribute?
- How do we find Information Gain?

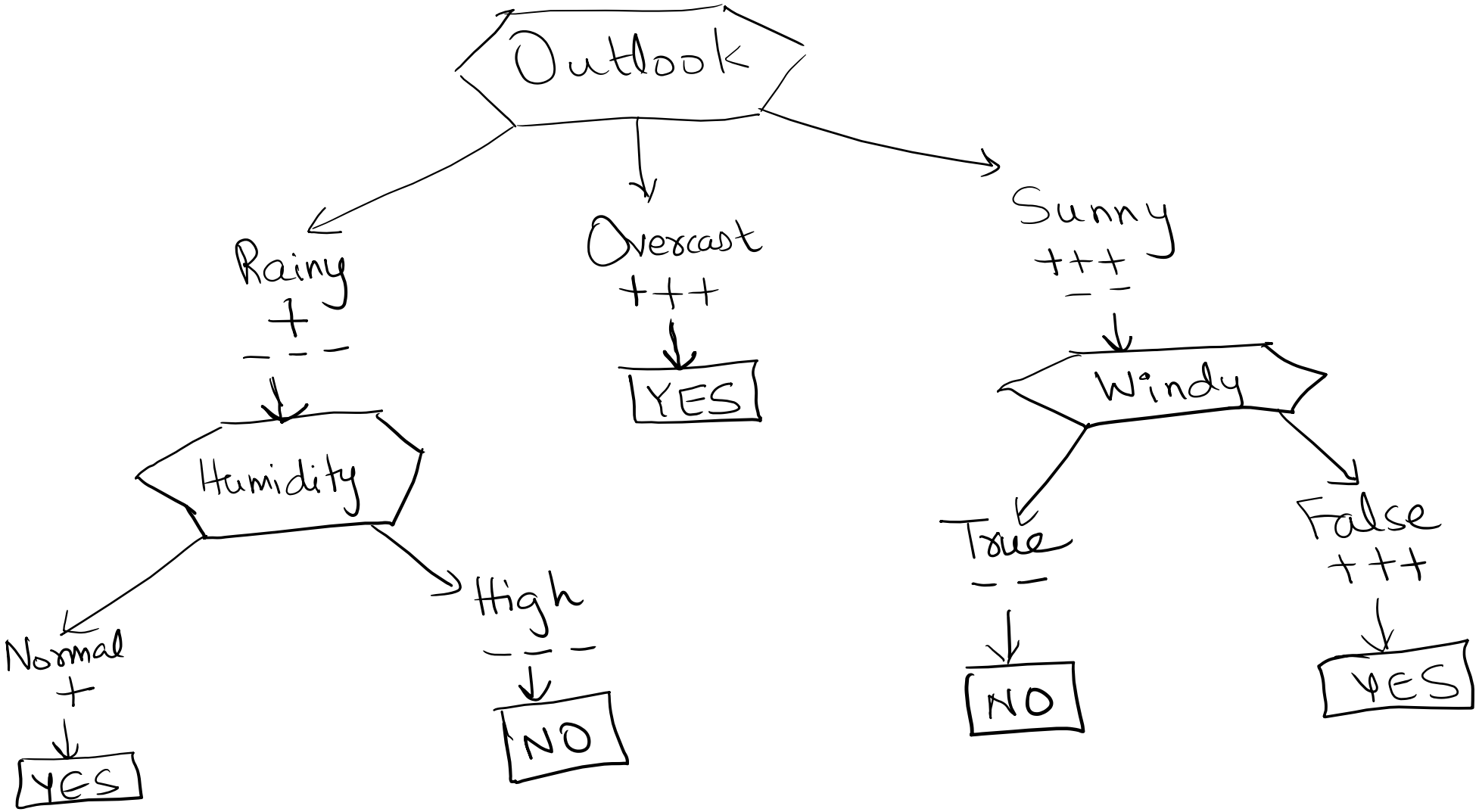
Training Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Hot	High	False	$y_1 = \text{No}$
$\boldsymbol{x}_2$	Rainy	Hot	High	True	$y_2 = \text{No}$
$\boldsymbol{x}_3$	Overcast	Hot	High	False	$y_3 = \text{Yes}$
$\boldsymbol{x}_4$	Sunny	Mild	High	False	$y_4 = \text{Yes}$
$\boldsymbol{x}_5$	Sunny	Cool	Normal	False	$y_5 = \text{Yes}$
$\boldsymbol{x}_6$	Sunny	Cool	Normal	True	$y_6 = \text{No}$
$\boldsymbol{x}_7$	Overcast	Cool	Normal	True	$y_7 = \text{Yes}$
$\boldsymbol{x}_8$	Rainy	Mild	High	False	$y_8 = \text{No}$
$\boldsymbol{x}_9$	Sunny	Mild	Normal	False	$y_9 = \text{Yes}$
$\boldsymbol{x}_{10}$	Rainy	Mild	Normal	True	$y_{10} = \text{Yes}$
$\boldsymbol{x}_{11}$	Overcast	Mild	High	True	$y_{11} = \text{Yes}$
$\boldsymbol{x}_{12}$	Sunny	Mild	High	True	$y_{12} = \text{No}$

Validation Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Cool	Normal	False	$y_1 = \text{Yes}$
$\boldsymbol{x}_2$	Overcast	Hot	High	True	$y_2 = \text{Yes}$

Final Decision tree



Stopping Criterion

- All samples in the node belong to same class
- No remaining attributes to split on
- Reduction in Impurity or information gain below a certain threshold
- Maximum Depth
- Minimum number of samples in node to split
- Minimum number of samples per leaf

Decision Trees

- What to do if after using all attributes we still have examples that do not belong to the same class?
- What to do if there are no examples in a particular subset?
- How to split if we have continuous valued attributes?
- How do we make decision boundaries with continuous valued attributes?

Overview of ID3 Algorithm

- ID3 (Iterative Dichotomiser 3) was developed by in 1986.
- Used for classification tasks only.
- Builds a tree using Information Gain from the Entropy measure.
- Constructs the tree top-down by selecting the best feature to split on.

How ID3 Works

- Calculate Entropy for the dataset.
- Compute Information Gain for each feature.
- Select the attribute with the highest Information Gain and split the dataset.
- Repeat recursively until all instances are classified or no features remain.

Advantages of ID3

- Simple and easy to implement.
- Finds the most informative attribute at each step.
- Works well for small datasets with categorical attributes.

Limitations of ID3

- Overfitting: ID3 grows a full tree without pruning.
- Handling of Continuous Data: Cannot handle numerical attributes directly.
- Bias toward Attributes with More Values: Prefers attributes with many unique values.

Introduction to C4.5 Algorithm

- C4.5 is an improved version of ID3, developed in 1993.
- Used for classification tasks only.
- Extends ID3 to handle continuous attributes, missing values, and pruning.
- Uses Gain Ratio instead of Information Gain.
- The Gain Ratio addresses ID3’s bias toward attributes with many unique values.

Gain Ratio

$$Gain\ Ratio(A) = \frac{Information\ Gain(A)}{Split\ Info(A)}$$

$$Split\ Info(A) = -\sum_{i=1}^{k} \frac{S_i}{S} \log_s \frac{S_i}{S}$$

Where k is the number of sets after split, S_i is the number of elements in set i and S is the total number of elements before split.

How C4.5 Works

- Handles Continuous Attributes by creating threshold-based splits.
- Computes Gain Ratio.
- Handles Missing Values using probability distributions.
- Applies Pruning to remove branches that do not contribute significantly to classification.

How C4.5 handles missing values

- During Split Selection:

$$Gain(A) = \frac{N_{known}}{N} \times Gain\ on\ known\ subset$$

- During Tree Construction:
  - Samples with missing attribute values are distributed fractionally

- During Prediction:
  - Sample with missing attribute goes through multiple branches and final prediction is a weighted sum of branch probabilities of different classes

Example

Student	Study Hours	Attendance (%)	Pass (1) or Fail (0)
A	1	50	0
B	5	80	1
C	6	85	1
D	2	40	0
E	3	60	0
F	8	90	1

63P, 37F = 100 values

$S \leq 4$

Yes

60 values = 27P, 33F

$A \leq 60$

Yes

6P, 24F

Fail

NO

21P, 9F

Pass.

NO

36P, 4F = 40 values.

Pass.

Comparison of ID3 and C4.5

- ID3 uses Information Gain, while C4.5 uses Gain Ratio.
- C4.5 handles numerical data, missing values, and pruning.
- ID3 is computationally simpler but less robust.

Advantages of C4.5

- Handles both categorical and continuous attributes.
- Deals with missing values effectively.
- Reduces overfitting through pruning.
- More robust than ID3.

Limitations of C4.5

- Computationally expensive: Gain Ratio calculations require more resources.
- Still prone to overfitting if not tuned properly.
- Bias toward attributes with high information gain.

Introduction to CART

- Stands for Classification and Regression Trees.
- CART is used for both classification and regression tasks.
- Uses binary recursive partitioning.
- Splits data into homogenous groups.

How CART Works

- Select the best feature to split.
- Split dataset into two child nodes.
- Recursively repeat until stopping criteria are met.
- Assign final predictions at leaf nodes.

Splitting Criteria for CART

- Gini Impurity: For Classification.
- Mean Squared Error (MSE): For Regression.
- Stopping Criteria: Min samples, max depth, impurity threshold, reduction in variance.

CART Algorithm for Regression

- Start with the entire dataset.
- Choose the best feature to split using MSE.
- Recursively split dataset into smaller binary subsets.
- Stop when a stopping criterion is met.
- Assign predictions as the mean value in each leaf node.

CART for Regression Example

Student	Study Hours	Attendance (%)	Marks
A	1	50	35
B	5	80	75
C	6	85	82
D	2	40	40
E	3	60	55
F	8	90	92

Questions?